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DBMS → DATABASE MANAGEMENT SYSTEM

ASSIGNMENT - 3

Section → A

Ques-1 What is SQL ? Why is it important in DBMS ?

SQL (Structured Query Language) is a standard language used to interact with a database. It is used to create, store, retrieve, update and delete data in a relational database.

It is important in DBMS →

1) Used to store and manage data efficiently.
2) Helps in retrieving specific data using queries.
3) Ensures data accuracy and consistency.

Ques-2 Difference between DDL and DML ?

Basis	DDL	DML
FULLFORM	Data Definition language	Data Manipulation language
PURPOSE	Defines structure of database	Manipulate data
COMMANDS	CREATE, ALTER, DROP	INSERT, UPDATE, DELETE
WORK ON	Tables, schema	Data (rows/records)
AUTO COMMIT	Yes (automatically saved)	NO (can rollback)
EXAMPLE	CREATE TABLE Student (id INT, name VARCHAR(50));	INSERT INTO Student VALUES (1,'Aisha')

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Ques-3 What is JOIN ? Explain It's Types.

A Join is used to combine data from two or more tables based on a common column (usually a foreign key).

It is needed because In databases, data is stored in multiple tables. JOIN helps to connect related data.

Types of JOIN →

(1) INNER JOIN → Returns only matching records from both tables.
(Only matching)

Ex →
```
SELECT Student.name, Department.dept-name
FROM Student
INNER JOIN Department
ON Student.dept_id = Department.dept-id;
```

(2) LEFT JOIN → Returns all records from left table + matched from right.
(All left + match)

Ex →
```
SELECT Student.name, Department.dept-name
FROM Student
LEFT JOIN Department
ON Student.dept-id = Department.dept-id;
```

(3) RIGHT JOIN → Returns all records from right table + matched from left
(All right + match)

Ex →
```
SELECT Student.name, Department.dept-name.
FROM Student
RIGHT JOIN Department
ON Student.dept-id = Department.dept-id;
```

(4) FULL JOIN → Returns all records from both tables.
(Everything)

```
SELECT Student.name, Department.dept-name
FROM Student
FULL JOIN Department
ON Student.dept-id = Department.dept-id;
```

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Ques-4 What is GROUP BY and HAVING clause?

GROUP BY → It is used to group rows having similar values in a column.

- Used with aggregate Fun^n like COUNT, SUM, AVG, MAX, MIN

SQL Query →
```
SELECT department, COUNT(*) AS total_students
FROM student
GROUP BY department;
```

HAVING → It is used to filter the grouped data based on a condition.

- Used after GROUP BY

SQL Query →
```
SELECT department, COUNT(*) AS total_students
FROM student
GROUP BY department
HAVING COUNT(*) > 1;
```

Ques-5 What is a subquery? Explain with example.

Subquery is also called Nested Query. It is a query written inside another SQL query. It is used to retrieve data that will be used by the main query.

Example Table →

Name:	Aisha	Rahul	Neha	Aman
marks:	85	70	90	60

Example → Find students with marks greater than average

```
SELECT name, marks
FROM student
WHERE marks > (
        SELECT AVG(marks) FROM student
);
```

Step by step, how it works →

1) Inner query runs first: SELECT AVG(marks) FROM Students (suppose average = 76)
2) Outer query becomes: SELECT name, marks FROM student WHERE marks > 76;

OUTPUT: Name: Aisha, Neha | marks: 85, 90

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Ques-6 What is Relational Algebra?

It is a procedural query language used in DBMS to perform operations on relations (tables) and retrieve desired results.

Basic operations in Relational Algebra →

(1) Selection (σ)

→ Selects rows based on condition.

Ex → σ marks > 80 (Student)

(2) Projection (π)

→ Selects specific columns

Ex → π names, marks (Student)

(3) Union (∪)

→ Combines two tables (no duplicate)

Ex → Student 1 ∪ Student 2

(4) Set Difference (−)

→ Returns records in one table but not in another

Ex → Student 1 − Student 2

(5) Cartesian Product (X)

→ Combines all rows of two tables.

Ex → Student X Department

(6) Join (⋈)

→ Combines tables based on condition

Ex → Student ⋈ Department

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Ques-7 Difference between WHERE and HAVING

WHERE → Used to filter rows before grouping.

HAVING → Used to filter groups after GROUP BY.

Basis	WHERE	HAVING
Purpose	Filters rows	Filters groups
Used with	SELECT	GROUP BY
Applied On	Individual rows	Grouped data
Aggregate Function	Not used	Used (COUNT, SUM etc)

SECTION → B

Ques-1 Write an SQL query to retrieve all students with marks > 75.

SQL Query →

```
SELECT *
FROM Student
WHERE marks > 75;
```

Explanation:

1) SELECT* → selects all columns from the table
2) FROM Student → Specifies the table name
3) WHERE marks > 75 → Filters only those students whose marks are greater than 75

The WHERE clause is used to apply conditions on rows.

Conclusion: This query helps in filtering & retrieving only required data using the WHERE clause.

Ans-1) Example:

id	name	marks
1	Aisha	85
2	Ram	70
3	Sneha	95

Result:

id	name	marks
1	Aisha	85
3	Sneha	95

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Ques-2 Write a query using JOIN to combine Student and course tables.

SQL Query →
```
SELECT Student.name, Course.course_name
FROM Student
INNER JOIN course
ON Student.course_id = Course.course_id;
```

Explanation →

1) INNER JOIN → returns only matching records from both tables
2) Student.course_id = Course.course_id → join condition
3) Combines student details with their course information.

Conclusion →

JOIN helps in combining related data from multiple tables using a common column.

Ques-3 Write a query using GROUP BY to count students per department.

SQL Query →
```
SELECT department, COUNT(*) AS total_students
FROM Student
GROUP BY department;
```

Explanation →

1) COUNT(*) → counts total number of students
2) GROUP BY department → groups data based on department
3) Each group shows number of students in that department.

Conclusion →

GROUP BY is used to group similar data and perform aggregation like counting.

Ans-2) Example Tables →

Student →

Id	Name	Course-id
1	Aish	101
2	Simmy	102

Course →

course-id	course-name
101	BCA
102	BBA

Result →

Name	course-name
Aish	BCA
Simmy	BBA

Ans-3) Example Table →

Id	Name	Department
1	Ashi	CS
2	Ram	IT
3	Sunny	CS
4	Diya	IT

Result →

Department	total-students
CS	2
IT	2

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Section - C

Ques-1 Find employees in a specific department using JOIN

To find employees working in a specific department, we can use JOIN along with a WHERE clause

SQL Query →

```
SELECT Employee.emp-name, Department.dept-name
FROM Employee
INNER JOIN Department
ON Employee.dept-id = Department.dept-id
WHERE Department.dept-name = 'HR';
```

Explanation →

1) INNER JOIN → combines Employee and Department tables
2) ON condition → links both tables using dept-id
3) WHERE clause → Filters employees of a specific department (HR)

Conclusion → JOIN helps to combine tables, and WHERE Filters data based on condition.

Ques 2 Retrieve total sales per customer.

To calculate total sales per customer, we use GROUP BY with SUM().

SQL QUERY →

```
SELECT customer_id, SUM(amount) AS total_sales
FROM orders
GROUP BY customer-id;
```

Explanation →

1) SUM(amount) → calculates total sales
2) GROUP BY customer-id → groups data per customer

Ans-1 Example Tables → Employee

emp-id	emp-name	dept-id
1	Aashi	101
2	Riya	102
3	Neelam	101

Department →

dept-id	dept-name
101	HR
102	IT

Result →

emp-name	dept-name
Aashi	HR
Neelam	HR

Ans-2 Example Table : Orders

order-id	customer-id	amount
1	101	500
2	102	300
3	101	200

Result →

customer-id	total-sales
101	700
102	300

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3) Each customer will have total purchase amount

Conclusion →
GROUP BY with SUM helps in calculating total values for grouped data.

Ques-3 A university database needs top 3 students. Write a query.
Based on marks, we use ORDER BY and LIMIT.
SQL Query →

```
SELECT name, marks
FROM Student
ORDER BY marks DESC
LIMIT 3;
```

Explanation →
1) ORDER BY marks DESC → sorts students in descending order
2) LIMIT 3 → Selects top 3 students
3) Returns highest scoring students

Conclusion →
ORDER BY and LIMIT help in retrieving top records from a dataset.

Ans- 3) Example Table →

Id	Name	marks
1	Aashi	85
2	Riya	70
3	Nisha	90
4	Aman	95

Result →

Name	marks
~~Aashi~~ Aman	95
Nisha	90
Aashi	85

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SECTION-D

Ques-1 Write an SQL query for any real-life system and explain logic.

Real-life system → E-Commerce (Online Shopping)

SQL Query →
```
SELECT customer-id, customer-name, amount
FROM orders
WHERE amount >1000;
```

Explanation → 1) SELECT customer-id, customer-name, amount
- Retrieves specific columns required for analysis.

2) FROM orders →
- ~~where~~ Refers to the table storing order details

3) WHERE amount >1000 →
- Filters only those orders where purchase value is greater than ₹1000.

Conclusion → SQL queries help extract meaningful information from large datasets for better decision-making.

Ques-2 Write a JOIN query and mention one use case.

SQL Query →
```
SELECT Student.name, Course.course-name
FROM Student
INNER JOIN Course
ON Student.course-id = Course.course-id;
```

Explanation → (1) Student Table contains student details
(2) Course Table contains course information
(3) INNER JOIN combines both tables
(4) ON condition matches course-id in both tables

Ans-1) Example Table: Orders

Order-id	customer-id	customer-name	amount
1	101	Siya	500
2	102	Janvi	1500
3	103	Simar	2000

Output →

customer-id	customer-name	amount
102	Janvi	1500
103	Simar	2000

Ans-2) Example Tables :

Student →

Id	Name	Course-id
1	Aashi	101
2	Riya	102

Course →

course-id	course-name
101	BCA
102	BBA

Output →

Name	Course-name
Aashi	BCA
Riya	BBA

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Conclusion →
JOIN is essential for combining related data from multiple tables and maintaining relationship.

Ques-3 Write a GROUP BY query and explain its output.

SQL Query →
SELECT department, COUNT(*) AS total_students
FROM student
GROUP BY department;

Explanation →
(1) GROUP BY department →
- Groups students based on department

(2) COUNT(*) →
- Counts number of students in each group

(3) AS total_students →
- Renames the column for clarity

Conclusion →
GROUP BY is used for grouping data and applying aggregate functions like COUNT, SUM, etc.

Ans-3) Example Table →

Id	Name	Department
1	Aisha	CS
2	Rahul	IT
3	Nisha	CS
4	Aman	IT

OUTPUT →

Department	total-students
CS	2
IT	2